

Površina ploskve

$M \subset \mathbb{R}^3$ ploskev s param. $\vec{r}: D \rightarrow \mathbb{R}^3$. Površina je

$$S(M) = \iint_M dS = \iint_D \|\vec{r}_u \times \vec{r}_v\| du dv = \iint_D \sqrt{EG - F^2} du dv$$

$$E = \|\vec{r}_u\|^2$$
$$G = \|\vec{r}_v\|^2$$
$$F = \langle \vec{r}_u, \vec{r}_v \rangle$$

13. Izračunaj površino tistega dela ploskve $x^2 + y^2 + z = 1$, ki leži nad ravnino xy .

$$z = 1 - x^2 - y^2$$

cil. koord.

$$z = 1 - r^2$$



→ eliptični paraboloid

param. $\vec{r} = \begin{bmatrix} r \cos \varphi \\ r \sin \varphi \\ 1 - r^2 \end{bmatrix}$

$$\vec{r}_r = \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ -2r \end{bmatrix}$$

$$\vec{r}_\varphi = \begin{bmatrix} -r \sin \varphi \\ r \cos \varphi \\ 0 \end{bmatrix}$$

$$r \in [0, 1]$$

$$\varphi \in [0, 2\pi]$$

$$E = \cos^2 \varphi + \sin^2 \varphi + 4r^2 = 4r^2 + 1$$

$$G = r^2(\sin^2 \varphi + \cos^2 \varphi) = r^2$$

$$F = -r \sin \varphi \cos \varphi + r \sin \varphi \cos \varphi + 0 = 0$$

$$\rightarrow S = \int_0^{2\pi} \int_0^1 \sqrt{(4r^2 + 1)r^2} dr d\varphi = 2\pi \int_0^1 r \sqrt{4r^2 + 1} dr = \frac{\pi}{4} \int_1^5 \sqrt{u} du = \frac{\pi}{4} \frac{2}{3} u^{3/2} \Big|_1^5 = \frac{\pi}{6} (5^{3/2} - 1)$$

$4r^2 + 1 = u$
 $8r dr = du$

Ploštni integral skal. polja

$$\iint_{\gamma} f dS = \iint_D f(\vec{r}) \|\vec{r}_u \times \vec{r}_v\| du dv = \iint_D f(\vec{r}) \sqrt{EG-F^2} du dv$$

(14.) Izračunaj m , težište in J_z kosa paraboida $z=x^2+y^2$, $z \leq 1$, $\delta = x^2+y^2$

cil. koord. $\rightarrow z=r^2, \delta=r^2$

$$\vec{r} = \begin{bmatrix} r \cos \varphi \\ r \sin \varphi \\ r^2 \end{bmatrix} \quad \vec{r}_r = \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ 2r \end{bmatrix} \quad \vec{r}_\varphi = \begin{bmatrix} -r \sin \varphi \\ r \cos \varphi \\ 0 \end{bmatrix} \rightarrow \sqrt{EG-F^2} = r\sqrt{4r^2+1}$$

$$m = \iint \delta dS = \int_0^{2\pi} \int_0^1 r^3 \sqrt{4r^2+1} dr d\varphi = 2\pi \frac{1}{8} \int_1^5 \sqrt{t} \frac{(t-1)}{4} dt = \frac{\pi}{16} \int_1^5 (t^{3/2} - t^{1/2}) dt =$$

$$\begin{aligned} 4r^2+1=t \\ 8r dr = dt \end{aligned} \quad = \frac{\pi}{16} \left(\frac{2}{5} (5^{5/2} - 1) - \frac{2}{3} (5^{3/2} - 1) \right) =$$

$$= \frac{\pi}{8} \left(5^{3/2} - \frac{1}{5} - \frac{1}{3} 5^{3/2} + \frac{1}{3} \right) =$$

iz simetrije $x^* = y^* = 0$

$$= \frac{\pi}{8} \left(\frac{2}{3} 5^{3/2} + \frac{2}{15} \right) = \frac{\pi}{12} \left(5^{3/2} + \frac{1}{5} \right)$$

$$m z^* = \int_0^{2\pi} \int_0^1 r^5 \sqrt{4r^2+1} dr d\varphi = 2\pi \frac{1}{8} \int_1^5 \sqrt{t} \frac{(t-1)^2}{16} dt = \frac{\pi}{64} \int_1^5 (t^{5/2} - 2t^{3/2} + t^{1/2}) dt =$$

$$= \frac{\pi}{64} \left(\frac{2}{7} (5^{7/2} - 1) - \frac{4}{5} (5^{5/2} - 1) + \frac{2}{3} (5^{3/2} - 1) \right) = \frac{\pi}{32} \left(\left(\frac{2}{5} - \frac{1}{7} - \frac{1}{3} \right) + 5^{3/2} \left(\frac{1}{7} 5^2 - \frac{2}{5} 5 + \frac{1}{3} \right) \right) =$$

$$= \frac{\pi}{4} \left(-\frac{1}{105} + 5^{3/2} \frac{5}{21} \right)$$

$$z^* = \frac{\frac{\pi}{4} \cdot \frac{5^{7/2}-1}{105}}{\frac{\pi}{12} \cdot \frac{1+5^{5/2}}{5}} = \frac{5^{7/2}-1}{7(5^{5/2}+1)} \rightarrow R^* = \left(0, 0, \frac{5^{7/2}-1}{7(5^{5/2}+1)} \right)$$

$$\frac{42-15-35}{21 \cdot 5} = \frac{-8}{105}$$

$$\frac{25 \cdot 3 - 42 + 7}{7 \cdot 3} = \frac{40}{21}$$

$$J_z = m z^* = \frac{\pi}{4} \left(\frac{5^{7/2}-1}{105} \right) = \frac{\pi(5^{7/2}-1)}{420}$$

Ploskovni integral vekt. polja

-orientirana ploskev M

$$\iint_M \vec{F} \cdot d\vec{S} = \iint_M \vec{F} \cdot \vec{N} ds = \int_0^1 \int_0^{2\pi} \langle \vec{F}(\vec{r}), \vec{r}_u \times \vec{r}_v \rangle du dv$$

16. Izračunaj pretok polja $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ skozi ploskev $z^2 = x^2 + y^2$, $0 \leq z < 1$ v smeri navzgor

$$\vec{r} = \begin{bmatrix} r \cos \varphi \\ r \sin \varphi \\ r \end{bmatrix} \quad \vec{r}_r = \begin{bmatrix} \cos \varphi \\ \sin \varphi \\ 1 \end{bmatrix} \quad \vec{r}_\varphi = \begin{bmatrix} -r \sin \varphi \\ r \cos \varphi \\ 0 \end{bmatrix} \quad \begin{array}{l} r \in [0, 1) \\ \varphi \in [0, 2\pi) \end{array}$$

$$\vec{F}(\vec{r}) = \begin{bmatrix} r \cos \varphi \\ r \sin \varphi \\ 2r \end{bmatrix} \quad \vec{r}_r \times \vec{r}_\varphi = \begin{vmatrix} i & j & k \\ \cos \varphi & \sin \varphi & 1 \\ -r \sin \varphi & r \cos \varphi & 0 \end{vmatrix} = \begin{bmatrix} -r \cos \varphi \\ -r \sin \varphi \\ r \end{bmatrix}$$

→ če želimo, da kaže navzdol vzamemo

$$\vec{r}_\varphi \times \vec{r}_r = -\vec{r}_r \times \vec{r}_\varphi$$

$$\rightarrow \iint \vec{F} \cdot d\vec{S} = \int_0^{2\pi} \int_0^1 \langle \vec{F}, \vec{r}_\varphi \times \vec{r}_r \rangle dr d\varphi = \int_0^{2\pi} \int_0^1 (r^2 - 2r^2) dr d\varphi = \underline{\underline{-\frac{2\pi}{3}}}$$

VEKT. DIF. OPERATORJI

1) skal. polje $\rightarrow$ vekt. polje : gradient

$$\vec{\nabla} = \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix}$$

2) vekt. polje $\rightarrow$ vekt. polje : rotor

$$\text{rot } \vec{F} = \vec{\nabla} \times \vec{F}$$

3) vekt. polje $\rightarrow$ skal. polje : divergenca

$$\text{div } \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial x} + \frac{\partial F_3}{\partial z}$$

Kompleks operatorjev:

• kompozicija dveh zap. operatorjev = 0

$$\text{rot} \circ \text{grad} = 0$$

$$\text{div} \circ \text{rot} = 0$$

$$\text{div} \circ \text{grad} = \Delta \text{ Laplaceov op.}$$

Vekt. polja ($\vec{F}$) so lahko

• potencialna: $\vec{F} = \text{grad } u$ za neki u

• irrotacionalna: $\text{rot } \vec{F} = 0$
(brez vrtincev)

• solenoidalna: $\text{div } \vec{F} = 0$
(brez izvirov)

• vsako pot. polje je irrotacionalno

• vsako polje, ki je rotor nekega polja, je solenoidalno.

• na dovolj lepih območjih (npr. zvezdastih, oz. brez lakenj) velja tudi obratno

18. $\vec{F} = \begin{bmatrix} x^2z \\ -xy^2z \\ 3yz^2 \end{bmatrix}, \text{div } \vec{F} = ?$

$$\text{div } \vec{F} = 2xz - 2xyz + 6yz$$